Observational Coordinates and Estimates: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Hour angle. Return →

$H = 8 - 6 = 2$ h, or 30°. Positive hour angle means the star is west of the meridian. **Check:** LST greater than RA means the object has already crossed the meridian.

Solution 2 — Meridian altitude. Return →

At upper culmination, $h = 90^\circ - |48^\circ - 20^\circ| = 62^\circ$. **Check:** the star is south of the zenith because its declination is less than the latitude.

Solution 3 — Altitude at hour angle. Return →

Use $\sin h = \sin 48^\circ \sin 20^\circ + \cos 48^\circ \cos 20^\circ \cos 30^\circ$. Numerically, $\sin h = 0.254 + 0.545 = 0.799$, so $h = 53.0^\circ$. **Check:** the altitude is lower than the meridian value of $62°$, as expected for $H = 30^\circ$.

Solution 4 — Airmass. Return →

At 30°, $X = 1/\sin 30^\circ = 2.0$. At 60°, $X = 1/\sin 60^\circ = 1.15$. **Check:** airmass grows rapidly near the horizon.

Solution 5 — Circumpolar limit. Return →

The limit is $\delta > 90^\circ - \phi = 42^\circ$. Stars north of +42° are circumpolar at latitude 48° N. **Check:** Polaris, near +89°, is circumpolar from almost anywhere in the northern hemisphere.

Solution 6 — Time above horizon. Return →

$\cos H_0 = -\tan 48^\circ \tan 20^\circ = -0.404$, so $H_0 = 113.8^\circ$. The time above the horizon is $2H_0/15 = 15.2$ h. **Check:** a northern star is above the horizon for more than half a day from a northern site.

Solution 7 — Zenith distance. Return →

$z = 90^\circ - 70^\circ = 20^\circ$. The airmass is $1/\sin 70^\circ = 1.06$. **Check:** near the zenith, airmass is close to 1.

Solution 8 — Sidereal drift. Return →

Stars transit about 3 min 56 s earlier each solar day, so after 30 days the transit is about 1 h 58 min earlier, near 20:02. **Check:** this is why the night sky changes with the seasons.

Solution 9 — Azimuth at culmination. Return →

A star with declination +20° transits due south, azimuth 180°, because its declination is less than the latitude. A star with declination +60° transits due north, azimuth 0°, because it is north of the zenith. **Check:** the zenith has declination equal to the latitude.

Solution 10 — Observing window. Return →

The hour angle is $20 - 18 = 2$ h, or 30° west of the meridian. Using $\sin h = \sin\phi \sin\delta + \cos\phi \cos\delta \cos H$ gives $h = 60.86^\circ$. A quadrant-aware azimuth formula gives $A = 242.8^\circ$ (west-southwest), and the Kasten–Young airmass is $X = 1.145$. The target is well placed, though past transit. **Check:** high altitude and small airmass make this a good observing window.